

flea

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FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

R. L. Bossard, Ph.D.
Editor, Flea News
bossardtech@gmail.com

Organizers of the Flea News Network are Drs. R. L. Bossard and N. C. Hinkle.

N. C. Hinkle, Ph.D.
Dept. of Entomology
Univ. of Georgia
Athens GA 30602-2603 USA
NHinkle@uga.edu
(706) 583-8043

Assistant Editor J. R. Kucera, M.S.

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Editorial

Keystone Fleas

Dear *Flea News* Reader,

Fleas in ancient times were regarded with amusement. As science showed fleas' propensity for pathogenicity, they became things to eradicate. Currently, there are draconian proposals to eradicate malaria or even entire taxa of parasites, such as mosquitoes, from the earth, but the ecological ramifications are of grave concern (Small 2019).

Can fleas and other parasites be "keystone species" in ecosystems? Yes, the evidence shows they can.

The keystone species concept was first demonstrated by Paine in 1969 when he removed marine predators experimentally. Community diversity declined as highly competitive species, otherwise regulated by the predators, monopolized resources. Keystone species regulate host populations, maintain community diversity, alter landscapes and life zones, and affect ecosystem function.

The idea that parasites and their microbial symbiotes could be keystone species, too, is a comparatively new one (Hatcher et al. 2014). Parasites comprise little biomass in the overall ecosystem, and are ignored in traditional ecosystem studies (Pojmariska 2002, Hudson et al. 2006).

Fleas are well known as carriers of a variety of other species (Durden and Hinkle 2019). These symbioses may be of widespread significance in regulating the functioning of terrestrial ecosystems.

Well wishes in regard to the coronavirus pandemic.

Yours in fleas,

R.L. Bossard

Editor, *Flea News*

Literature

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Announcements

2020 Biology of Vector-borne Diseases Course

The Center for Health in the Human Ecosystem hosts the annual Biology of Vector-borne Diseases six-day course. The course is scheduled for June 21-26, 2020, on the University of Idaho campus in Moscow, Idaho. If you haven't been to Idaho in June, it's a beautiful time of year to visit.

We're looking forward to another terrific year with some exciting new additions to the course. We've engaged some new instructors along with our returning core (the 2020 instructor list is posted online), including our highlight presentation this year on Friday by Robert Ricklefs, a National Academy of Sciences member and Professor at the University of Missouri-St. Louis, who will be speaking on his career-long work on avian malaria and its impact on ecology and evolution of global bird populations.

MISSION: Create a knowledge network for a diverse community of practitioners that persists, grows and transforms science, and interventions for plant, animal and human vector-borne diseases.

This course provides accessible, condensed training and "knowledge networking" for advanced graduate students, postdoctoral fellows, faculty and professionals to ensure competency in basic biology, current trends and developments, and practical knowledge for U.S. and global vector-borne diseases of plants, animals and humans. We seek to train the next generation of scientists and help working professionals to more effectively address current and emerging threats with holistic approaches and a strong network of collaborators and mentors.

www.uidaho.edu/cals/center-for-health-in-the-human-ecosystem/education/vector-borne-diseases

Shirley Luckhart and Ed Lewis
Department of Entomology, Plant Pathology and Nematology
875 Perimeter Drive MS 2329
University of Idaho
Moscow, ID 83844-2329

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Livestock Insect Workers Conference 2020

The LIWC conference dates are June 14-17, with our typical schedule Sunday to Wednesday morning. The hotel will be the Omni Dallas at Park West, 1590 Lyndon B Johnson Fwy, Dallas, TX 75234. A call for abstracts and registration info will be sent out closer to the meeting.

<https://www.liwconline.org/welcome>

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Society for Vector Ecology (SOVE)

Annual Conference Information

SOVE 2020 scheduled for September 20-24 in Honolulu, Hawaii.

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Updated contact for Dr. Norbert Mencke

Grunder Muehle 2 B, 51381 Leverkusen, Germany. mencke_norbert@yahoo.com

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Featured Research

New Flea Species 2019

Acosta, R., & Salceda-Sánchez, B. (2019). A new species of flea of the genus *Kohlsia* Traub, 1950 (Siphonaptera: Ceratophyllidae) from Veracruz, Mexico. *Zootaxa*, 4671(4), 564-570. doi:10.11646/zootaxa.4671.4.7

[*Kohlsia misantlensis* n. sp.]

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Jean-Claude Beaucournu, Vincent Prié & Anne Laudisoit. 2019. Le genre *Ctenophthalmus* Kolenati, 1856 (Siphonaptera : Ctenophthalmidae) en région afrotropicale; description d'un nouveau taxon; répartition et spécificité des sous-genres *Ethioctenophthalmus*, *Geoctenophthalmus* et *Idioctenophthalmus*. *Annales Soc. Entom. France* (N S), 2019, 55: 355-382. <https://doi.org/10.1080/00379271.2019.1637276>.

The genus *Ctenophthalmus* Kolenati, 1856 (Siphonaptera: Ctenophthalmidae) in the Afrotropical region. Description of a new taxon; distribution and specificity of the subgenera *Ethioctenophthalmus*, *Geoctenophthalmus* and *Idioctenophthalmus*. As vectors of the plague bacillus (*Yersinia pestis*) and other neglected agents (*Rickettsia*, *Bartonella*) fleas of small mammals should, in endemic areas, be inventoried, their distribution documented, and their specificity as well as their vector capacity assessed. Such is partly the objective of the present work, which attempts to update the list of species of a particularly abundant and widely distributed flea genus in the afrotropical region, the genus *Ctenophthalmus*. The present work summarizes available knowledge for the genus *Ctenophthalmus* in Africa, presents some new data and corrects erroneous data from the literature. A new subspecies, *Ctenophthalmus* (*Ethioctenophthalmus*) *calceatus obesus* Beaucournu & Prié n. ssp. is described from Burundi based on material collected by C. Guiguen et J. Vissault in 1978. The different taxa (76 species and subspecies) are grouped into informal groups ("species groups" initially defined by Smit in 1963) and we propose a new group, the *cophurus* group. We recall for each taxon the available data on type locality and type material, type host, known distribution and other known hosts. For the latter, a special attention has been paid to recent taxonomic changes and to the hosts distribution. A map has been produced for each taxa. The distribution we observe, taxon by taxon, does not correspond to this day to that of these Siphonaptera, but rather to that of the researchers efforts. Specificity—despite an obvious bias linked to intensive trapping effort in the context of eco-epidemiological studies—shows the preponderance of Muridae as hosts, but the potential importance of small burrowing mammals (rodents and insectivores) in *Ctenophthalmus* occurrence is poorly known. A better knowledge of the species and their distribution should help a better understanding of the sylvatic plague in Africa.

[*Ctenophthalmus* (*Ethioctenophthalmus*) *calceatus obesus* Beaucournu & Prié n. ssp.]

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Keskin, A., Beaucournu, J.-C. 2019. Descriptions of Two New Species and a New Subspecies of the Genus *Ctenophthalmus* (Insecta: Siphonaptera: Ctenophthalmidae) from Turkey. (2019) *Journal of medical entomology*, 56 (5), pp. 1275-1282.

Two new species and a new subspecies of fleas belonging to the genus *Ctenophthalmus*

Kolenati (Siphonaptera: Ctenophthalmidae) were described from Turkey. *Ctenophthalmus (Spalacoctenophthalmus) beyzanuræ* n. sp. was collected from *Nannospalax leucodon* Nordmann (Mammalia: Rodentia) in Tekirdağ province while *Ctenophthalmus (Euctenophthalmus) kefelioglu* n. sp. and *Ctenophthalmus (Euctenophthalmus) teres anaticus* n. ssp. were collected from *Microtus guentheri* (Danford & Alston) and *Apodemus uralensis* (Pallas) (Mammalia: Rodentia) in the Aksaray and Kars provinces of Turkey, respectively. Detailed illustrations of new taxa described were also provided.

[*Ctenophthalmus (Spalacoctenophthalmus) beyzanuræ* n. sp.

Ctenophthalmus (Euctenophthalmus) kefelioglu n. sp.

Ctenophthalmus (Euctenophthalmus) teres anaticus n. ssp.]

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Keskin, A., Beaucournu, J.-C. 2019. *Palaeopsylla (Palaeopsylla) aysenuræ* n. sp., a new ctenophthalmid flea (Siphonaptera: Ctenophthalmidae) from Turkey. (2019) Zootaxa, 4613 (2), pp. 369-374.

A new flea species, *Palaeopsylla (Palaeopsylla) aysenuræ* n. sp., belonging to the *Palaeopsylla* minor-group is described and illustrated. Specimens (seven males and nine females) of the new species were collected from *Talpa levantis* Thomas, 1906 (Mammalia: Talpidae) from Trabzon province of Turkey. An identification key to all known species of the *Palaeopsylla* minor-group is provided.

[*Palaeopsylla (Palaeopsylla) aysenuræ* n. sp.]

Taxonomy

Roe, D. 2019. Biodiversity loss—more than an environmental emergency. The Lancet: Planetary Health Volume 3, ISSUE 7, Pe287-e289, July 01, 2019. DOI:[https://doi.org/10.1016/S2542-5196\(19\)30113-5](https://doi.org/10.1016/S2542-5196(19)30113-5)

"Although some loss of biodiversity is normal, the current rate of extinction is unprecedented being up to 1000 times higher than natural background rates. Development professionals need to engage, because the current rate of loss poses a severe threat to key development priorities (eg, health and hunger)."

LOPEZ BERRIZBEITIA M. F; J. P. SANCHEZ; R. BARQUEZ & M. DIAZ. "TAXONOMIC REVISION OF THE FLEA GENUS *AGASTOPSYLLA* JORDAN AND ROTHSCILD 1923 (SIPHONAPTERA: CTENOPHTHALMIDAE)". ANNALS OF THE BRAZILIAN ACADEMY OF SCIENCES (IN PRESS).

Fleas of Argentina are receiving renewed systematic interest, but the identification of many species associated with small mammals can be problematic. We review the taxonomy of the flea genus *Agastopsylla* and include the re-description of two species and one subspecies, and designate a neotype and neallotype for *Agastopsylla hirsutior* and a neotype for *Agastopsylla nylota nylota* from the Colección Mamíferos Lillo Anexos (CMLA), Universidad Nacional de Tucumán, Argentina. Additionally, a key to identification of the species of *Agastopsylla* and a distribution map of the species of the genus are presented.

Hosts and Distribution

Urdapilleta M., Linardi P.M. & Lareschi, M. 2019. Fleas associated with sigmodontine rodents and marsupials from the Paranaense Forest in Northeastern Argentina. *Acta Tropica* 193: 71-77, <https://doi.org/10.1016/j.actatropica.2019.02.011>

Zurita A, R. Callejón, M. Urdapilleta, M. Lareschi & C. Cutillas. 2019. Origin, Evolution, Phylogeny and Taxonomy of *Pulex irritans* (Siphonaptera: Pulicidae). *Medical and Veterinary Entomology* 33: 296-311. doi: 10.1111/mve.12365.

Lareschi M., Savchenko E. & M. Urdapilleta. 2019. Ectoparasites associated with sigmodontine rodents from northeastern Argentina. *Therya* 10 (2) 103-108. DOI: 10.12933/therya-19-758 ISSN 2007-3364.

Orozco-Valor, P.M., Santillan M., M. Lareschi & J.M. Grande. First record of the flea *Polygenis platensis* parasitizing the free-living American Kestrels. *Journal of Raptor Research* 53 (4). In press

Moreno Salas L., M. Espinoza-Carniglia, N. L. Schmeisser, L. G. Torres Fuentes, C. Silva-De La Fuente, M. Lareschi, D. González-Acuña. 2019. Fleas Of Black Rats (*Rattus rattus*) as reservoir host of *Bartonella* spp. in Chile. *PeerJ* 7:e7371 <https://doi.org/10.7717/peerj.7371>

Savchenko E & M. Lareschi. 2019. Molecular and morphological description of a new species of *Laelaps* Koch, 1836 (Mesostigmata: Laelapidae) parasitic of *Oligoryzomys flavescens* Waterhouse (Rodentia: Cricetidae) from Argentina. *Acta Tropica* 199: 105-146

Urdapilleta M., Cicuttin GL, De Salvo MN, Pech May A, Salomon OD & Lareschi, M. Molecular detection and identification of *Bartonella* in the cat flea *Ctenocephalides felis felis* collected from cats and dogs in a border area in northeastern Argentina. *Veterinary Parasitology: Regional Studies and Reports* 19 (2020): 100361 <https://doi.org/10.1016/j.vprsr.2019.100361>.

Melis M., Savchenko E., Ferrari W., Galliari C., Nava S., Lareschi M. Caracterización de la comunidad ectoparásita de roedores sigmodontinos (Cricetidae) en un ámbito hortícola bonaerense; prospección de *Rickettsia* spp. VIII Congreso Argentino de Parasitología. Corrientes, del 24 al 27 de abril de 2019. *Revista de la Asociación Parasitológica Argentina*. Número especial. Pag 100

Urdapilleta, M.; Burgos E. F.; Lamattina D.; Lareschi M.; Salomón O. D. Animales de compañía y sinantrópicos como reservorios de la pulga *Tunga penetrans* en barrios afectados de la ciudadde Puerto Iguazú, Misiones. VIII Congreso Argentino de Parasitología. Corrientes, del 24 al 27 de abril de 2019. Revista de la Asociación Parasitológica Argentina. Número especial. Pag. 107

Lizama-Schmeisser, N., Espinoza-Carniglia, M., Lareschi, M., Gonzalo-Fuentes, L., González Acuña, D. & Moreno-Salas, L. Actualización de pulgas (Orden Siphonaptera) asociadas a roedores en Chile. XL Congreso Nacional de Entomología, Sociedad Entomológica de Chile. Concepción, Chile, del 2 al 4 de octubre de 2019. Pág. 105

Haas, G. E., Johnson, L., & Wilson, N. (2019). Siphonaptera from mammals in Alaska. Supplement II. Southeastern Alaska. Journal of the Entomological Society of British Columbia, 77, 43-46.

Haas, G. E., Johnson, L., & Wood, R. E. (2019). Siphonaptera from mammals in Alaska. Supplement IV. Revised check list for Southeastern Alaska. Journal of the Entomological Society of British Columbia, 79, 54-61.

Hamidi, K., & Nassirkhani, M. (2019). Annotated checklist of fleas (Insecta: Siphonaptera) and lice (Insecta: Anoplura) associated with rodents in Iran, with new reports of fleas and lice. Journal of Vector Borne Diseases, 56(2), 134.

Physiology

Kakumanu, R., Hodgson, W.C., Ravi, R., Alagon, A., Harris, R.J., Brust, A., Alewood, P.F., Kemp-Harper, B.K., Fry, B.G. 2019. Vampire venom: Vasodilatory mechanisms of vampire bat (*Desmodus rotundus*) blood feeding. (2019) Toxins, 11 (1), art. no. 26.

Animals that specialise in blood feeding have particular challenges in obtaining their meal, whereby they impair blood hemostasis by promoting anticoagulation and vasodilation in order to facilitate feeding. These convergent selection pressures have been studied in a number of lineages, ranging from fleas to leeches. However, the vampire bat (*Desmondus rotundus*) is unstudied in regards to potential vasodilatory mechanisms of their feeding secretions (which are a type of venom). This is despite the intense investigations of their anticoagulant properties which have demonstrated that *D. rotundus* venom contains strong anticoagulant and proteolytic activities which delay the formation of blood clots and interfere with the blood coagulation cascade. In this study, we identified and tested a compound from *D. rotundus* venom that is similar in size and amino acid sequence to human calcitonin gene-related peptide (CGRP) which has potent vasodilatory properties. We found that the vampire bat-derived form of CGRP (i.e., vCGRP) selectively caused endothelium-independent relaxation of pre-contracted rat small mesenteric arteries. The vasorelaxant efficacy and potency of vCGRP were similar to that of CGRP, in activating CGRP receptors and Kv channels to relax arteriole smooth muscle, which would facilitate blood meal feeding by promoting continual blood flow. Our results provide, for the first time, a detailed investigation into the identification and function of a vasodilatory peptide found in *D. rotundus* venom, which provides a basis in understanding the convergent pathways and selectivity of hematophagous venoms. These unique peptides also show excellent drug design and development potential, thus highlighting the social and economic value of venomous animals.

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Roth, J.D., Stephen Dobson, F., Criscuolo, F., Uhlich, P., Zahariev, A., Bergouignan, A., Viblanc, V.A. 2019. Subtle short-term physiological costs of an experimental augmentation of fleas in wild Columbian ground squirrels. *Journal of Experimental Biology*, 222 (12), art. no. jeb203588.

Parasites affect many aspects of host physiology and behavior, and thus are generally thought to negatively impact host fitness. However, changes in form of short-term parasite effects on host physiological markers have generally been overlooked in favor of fitness measures. Here, we studied flea (*Oropsylla idahoensis* and *Oropsylla opisocroistis tuberculata*) parasitism on a natural population of Columbian ground squirrels (*Urocitellus columbianus*) in Sheep River Provincial Park, AB, Canada. Fleas were experimentally added to adult female *U. columbianus* at physiologically demanding times, including birth, lactation and weaning of their young. The body mass of adult females, as well as their oxidative stress and immunity were recorded multiple times over the active season under flea-augmented and control conditions. We also measured the prevalence of an internal parasite (*Trypanosoma otospermophili*). Doubly labeled water (DLW) was intra-peritoneally injected at peak lactation to examine energy expenditure. Effects of parasites on oxidative stress were only observed after offspring were weaned. There was no direct effect of experimentally heightened flea prevalence on energy use. A short-term 24 h mass loss (−17 g) was detected briefly after parasite addition, likely due to *U. columbianus* preferentially allocating time for grooming. Our parasite augmentation did not strongly affect hosts and suggested that short-term physiological effects were unlikely to culminate in long-term fitness consequences. Columbian ground squirrels appear to rapidly manage parasite costs, probably through grooming.

Ecology

Small, E. In defence of the world's most reviled invertebrate 'bugs'. *Biodiversity Volume 20*, 2019 - Issue 4 : 168-221.

Species of invertebrate animals, notably insects, are undergoing an alarmingly high rate of extinction, coupled with minimal support for their protection, even from the world's leading conservation organisations. This is intolerable, as invertebrates constitute over 95% of the world's species, have indispensable economic values and provide ecological services without which life on earth would virtually cease. Much of the lack of public and governmental support for invertebrate conservation is due to the abhorrent tiny pests that have persuaded most people that 'bugs' are bad and consequently the only species worthy of support are the charismatic superstar mammals like pandas and tigers that currently are the mainstays of biodiversity fundraising. Just as these respected, highly attractive icons are effective ambassadors of biodiversity conservation, so certain detested pests have poisoned the public image of invertebrates, and indeed have made it seem to many that most wildlife is hostile. The 'dirty dozen' bugs that particularly are a hindrance to improving public investment in biodiversity are: bedbugs, clothes moths, cockroaches, fleas, houseflies, leeches, lice, locusts, mosquitoes, spiders, termites and ticks. Except for spiders, these species, admittedly, are responsible for enormous damage to health and economic welfare. Nevertheless, this paper shows that most have at least some compensating values, their harm has often been exaggerated and all have related species that are good citizens. Six of the dozen 'least wanted' invertebrates highlighted are blood parasites of

humans, and these ‘bad apples’ are very hard to defend since parasitism seems abhorrent. Remarkably, however, at least half of the world’s tens of millions of species are also parasites, and without them most ecosystems would be in danger of collapse. To improve invertebrate conservation, it is advisable that efforts be made to educate the public regarding their importance. Since prejudices against ‘bugs’ are primarily acquired during childhood, special attention is needed to persuade the young that most invertebrates are harmless, valuable and entertaining. Recent advances in genetic engineering (‘synthetic biology’, ‘genetic drives’) have led to very serious consideration of deliberately eliminating the world’s worst pests of humans. While these extermination technologies could greatly increase support for invertebrate conservation by annihilating their most despised representatives, the dangers of unforeseen damage to ecosystems and hence to biodiversity are substantial.

Alonso R., Ruiz M., Lovera R., Montes De Oca D. P., Cavia R. & J. P. Sánchez. 2019. Norway rat (*Rattus norvegicus*) ectoparasites in livestock production systems from central Argentina: influencing factors on parasitism. *Acta Trópica* (in press).

Haematophagous ectoparasites are worldwide vectors of many zoonotic bacterial diseases, both emerging and re-emerging, whose incidences are rising. Livestock development alters different environmental characteristics such as the microclimate of a site, changing the availability, density and susceptibility of the hosts to pathogens and vectors, indirectly influencing the spread and persistence of a disease within an ecosystem. The Norway rat (*Rattus norvegicus*), the most abundant vertebrate pest species found on livestock farms from Argentina, is a reservoir for several important zoonotic bacteria and may harbor ectoparasite species, which act as their vectors. Even though the Norway rat is widely known for its role as an ectoparasite host, the ecological characteristics of their ectoparasite communities and the related factors with parasitism on livestock farms have never been described. In the present study, we describe the ectoparasite community in Norway rats from central Argentina livestock farms, while also depicting the influencing factors on both ectoparasite occurrence and abundance. Ectoparasites were collected from rats captured in 20 sites from Buenos Aires province, between the winter of 2016 and the summer of 2018. A total of 1441 ectoparasite individuals were collected from 159 Norway rat individuals [Total ectoparasite prevalence = 69.2%; Mean ectoparasite specimen abundance ($\pm$ CI) = 9.06 ± 2.32 ectoparasite individuals per rat; Mean ectoparasite specimen intensity ($\pm$ CI) = 13.10 ± 3.08 ectoparasite individuals per infested rat found]. Ectoparasite assemblage consisted of four cosmopolitan species, recognized for their sanitary relevance: mites (Laelapidae: *Laelaps nuttalli* and *Laelaps echidninus*), lice (Polyplacidae: *Polyplax spinulosa*) and fleas (Pulicidae: *Xenopsylla cheopis*). We observed higher Norway rat abundance in sites related to higher ectoparasite occurrence and abundance frequencies on the rats. Additionally, ectoparasites were more abundant on rats in warm seasons and on male individuals, over female rats. Moreover, the geographical location of the studied sites influenced the ectoparasite assemblage structure observed on the rats. This study broadens the knowledge on the role of Norway rats as zoonotic ectoparasites hosts and analyzes the drivers influencing ectoparasite occurrence and abundance on the most populated region of Argentina, which is also the region with the most intensive livestock farming. Therefore, this survey may assist in evaluating potential risks for humans and generate effective sanitary control strategies for ectoparasite-borne infectious diseases.

Sánchez, J. P. & M. Lareschi. 2019. Diversity, distribution and parasitism rates of fleas (Insecta: Siphonaptera) on Sigmodontine rodents (Cricetidae) from Argentinian Patagonia. *Bulletin of*

Entomological Research, 109 (1), 72-83. <https://doi.org/10.1017/S0007485318000196>

Fleas have great medical relevance as vectors of the causative agents of several diseases in animals and humans and rodents are the principal reservoirs for these pathogens. Argentinian Patagonia has the highest diversity of rodent fleas in South America. However, parasitism rates of rodents by fleas, the factors that influence them and the ecological aspects that modulate geographical distributions of flea–host association remain unknown for this region. This is the first study to record the diversity, prevalence, abundance, geographical distributions and host ranges of fleas in Argentinian Patagonia. It also compares parasitism rates among Patagonian ecoregions and host species. We captured 438 rodents belonging to 13 species, which harboured 624 fleas from 11 species and subspecies ($P = 46\%$; mean abundance = 1.44). The high parasitism rates obtained were consistent with previous records for other arid regions, suggesting that Patagonia favours the survival and development of Siphonaptera. Host geographic range and abundance were related to the parasitological indexes: host species with high-density populations had the highest mean flea abundance and prevalence, whereas widely distributed hosts had the highest richness and diversity of flea species. Our results contribute to the knowledge of the flea–host–environment complex. Our analysis of flea distributions and parasitism rate in Central Patagonia may be useful in epidemiological studies of flea-borne diseases and provide a basis for implementing surveillance systems for better risk assessment of emerging zoonoses in the region.

Kluever, B.M., Iles, D.T., Gese, E.M. 2019. Ectoparasite burden influences the denning behavior of a small desert carnivore. (2019) *Ecosphere*, 10 (5), art. no. e02749. DOI: 10.1002/ecs2.2749

Quantifying the impacts of parasitism on a host can be arduous and is generally understudied for ectoparasites, with known works being either laboratory-focused, correlational-based, or only focusing on a few species and spatial extents. Many mammalian species have evolved the modality of denning behavior, a lifestyle that can lead to higher ectoparasite burden, and it has been posited that animals may alter their denning behavior in an attempt to reduce exposure to ectoparasites. We conducted a test of the ectoparasite release hypothesis for kit foxes (*Vulpes macrotis*) and fleas in the Great Basin Desert of the western United States, a hypothesis that has remained relatively untested for over half a century. We experimentally administered a flea reduction treatment to a subset of kit foxes. We then measured and compared the number of unique den usages and residency time across treatment and control foxes (no flea reduction treatment) while accounting for other factors known to influence denning behavior. Foxes treated with the flea medication reduced the number of unique dens and increased their residency times at dens. All kit foxes continued to use multiple dens on the landscape, suggesting several factors in addition to flea burden influence denning behavior. Our results confirm the long-dormant ectoparasite release hypothesis and suggest ectoparasites may shape the behavior of burrowing vertebrates to a greater extent than previously recognized.

"Flea species encountered and identified, in order of prevalence, were *Pulex irritans*, *Meringis parkeri*, *Foxella ignota utahensis*, *Aetheca wagneri*, and *Peromyscopsylla hesperomys adelpha*; these species were similar to previous flea investigations for kit fox and congener species (Bossard 2006, Riner et al. 2018)".

(Bossard, R. L. 2006. Mammal and flea relationships in the Great Basin Desert: from H. J. Egoscue's collections. *Journal of Parasitology* 92:260–266.

Riner, A. J., J. L. Rudd, D. L. Clifford, B. L. Cypher, J.E. Foley, and P. Foley. 2018. Comparison of flea (Siphonaptera) burdens on the endangered San Joaquin kit fox (*Vulpes macrotis mutica*) inhabiting urban and nonurban environments in Central Valley, California. *Journal of Medical Entomology* 55:995–1001.)

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Pathology and Control

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**

Directory of Siphonapterists (updated)

Dra. Roxana Acosta-Gutierrez
Museo de Zoología
Departamento de Biología Evolutiva
Facultad de Ciencias, UNAM
Apo. Postal 70-399,
C.P. 04510, Mexico. D.F.
roxana_a2003@yahoo.com.mx
Flea biogeography, systematics, and taxonomy.

Dr. J.C. Beaucournu
Parasitologie medicale
Faculte de Medecine de Rennes
2, avenue du Professeur Leon Bernard
F 35043 Rennes cedex
France
jc.beaucournu@gmail.com

Dr. Marian Blaski
Silesian University, Department of Zoology
ul. Bankowa 9
40-007 Katowice
Poland
marian.blaski@us.edu.pl

Dr. R.L. Bossard
Bossard Consulting
Salt Lake City, UT

bossardtech@gmail.com
Ecology of host-parasite relationships.

Dra. Cristina Cutillas
Departamento de Microbiología y Parasitología
Facultad de Farmacia
Universidad de Sevilla
C/ PROFESOR GARCÍA GONZÁLEZ, N° 2
41012 Sevilla. Spain
cutillas@us.es
Molecular biology of fleas

Dr. Anne Darries-Vallier
Bio Espace – Laboratoire d'Entomologie
Mas des 4 Piles - Route de Bel-Air
34570 Murviel les Montpellier
France
bioespace.labo@gmail.com
Ecology, biology, behavior, mass rearing, disease vectors.
Bill Donahue, Ph.D.
Sierra Research Laboratories
5100 Parker Road
Modesto, CA 95357
www.sierraresearchlaboratories.com
bill@sierraresearchlaboratories.com

Michael W Dryden DVM, PhD, DACVM
University Distinguished Professor of Veterinary Parasitology
Department of Diagnostic Medicine/Pathobiology
202 Trotter Hall
Manhattan, KS 66506
Dryden@vet.k-state.edu
Veterinary parasitology.

Thomas M. Dykstra, Ph.D.
Dykstra Laboratories, Inc.
3499 NW 97th Blvd., Suite 6
Gainesville, FL 32606
www.dykstralabs.com
dykstralabs@yahoo.com
Sensory perception of fleas, especially larvae.

Lance A. Durden, Ph.D.
Department of Biology
Georgia Southern University

69 Georgia Avenue
 P. O. Box 8042
 Statesboro, Georgia 30460, USA
 ldurden@georgiasouthern.edu
 Flea taxonomy and host associations, especially in eastern North America and the Indo-Australian region.

Laura Fielden (Ph.D.)
 Associate Professor, Department of Biology
 Truman State University, Kirksville, MO 63501
 lfielden@truman.edu
 Host specificity of fleas.

Manuel Fabio Flechoso del Cueto
 C/ Heroes de la Independencia no 1 - 2oA
 42200 Almazan (Soria), SPAIN
 fabioflechoso@hotmail.com

Dr. Patrick Foley
 Department of Biological Sciences
 California State University
 Sacramento, CA 95819
 patfoley@csus.edu
 Epidemiology, extinction, metapopulations, identification of fleas, plague.

Dr. Terry Galloway
 Professor of Entomology and Associate Curator
 J.B. Wallis Museum of Department of Entomology
 Winnipeg, Manitoba
 Canada R3T 2N2
 Terry_Galloway@umanitoba.ca
 Ecology and taxonomy, especially of the larvae.

Dr. N. C. Hinkle
 Professor
 Dept. of Entomology
 Univ. of Georgia
 Athens, GA 30602-2603
 NHinkle@uga.edu
 Flea biology and control.

Simon Horsnall
 UK Flea Recorder
 simon.horsnall@googlemail.com

Kerv Hyland
 5 Timber Lane, Unit 314
 Exeter, N.H. 03833

khyland@etal.uri.edu

All groups of ectoparasites on vertebrates.

James (Jim) R. Kucera, M.S.

5930 Sultan Circle

Murray, UT 84107-6930

jameskucera@aol.com

Flea systematics & taxonomy, host relationships, distribution & biogeography.

Dra. Marcela Lareschi

Centro de Estudios Parasit. Vectores

CEPAVE (CONICET-UNLP)

Calle 2 # 584

La Plata (1900)

Argentina

mldareschi@cepave.edu.ar

Dr. Pedro Marcos Linardi

Professor of Parasitology and Medical Entomology

Departamento de Parasitologia

Instituto de Ciencias Biologicas

Universidade Federal de Minas Gerais

31.270-901 Belo Horizonte, Minas Gerais

Brasil

linardi@icb.ufmg.br

Taxonomy of fleas, host-parasite relationships, fleas as vectors of parasites.

Dr. Erica McAlister

Department of Life Sciences

The Natural History Museum

Cromwell Road LONDON SW7 5BD UK

Fleas at the NHM.

Christine M. McCoy, M.S.

Staff Entomologist

BerTek, Inc.

Greenbrier, AR 72058

cmccoy@bertekconsults.com

Sergei G. Medvedev, Doctor of Biology

Chief of the Department of Parasitology

Zoological Institute of Russian Academy of Sciences

Universitetskaya Embankment 1
 St. Petersburg, 199034 Russia
<http://www.zin.ru/Animalia/Siphonaptera/index.htm>
fleas@zin.ru
 Flea taxonomy and phylogenetics.

Dr. Norbert Mencke
 Grunder Muehle 2 B
 51381 Leverkusen
 Germany
mencke_norbert@yahoo.com
 Veterinary specialist for parasitology.

Ettore Napoli
 Department of Veterinary Science unit Parasitology
 University of Messina
 Messina, Italy
enapoli@unime.it
 Ectoparasitic arthropods and their vectorial role.
 MDV PhD student - Public Health

Dr. Barry M. Oconnor, Curator & Professor
 Museum of Zoology, University of Michigan
 1109 Geddes Ave
 Ann Arbor, MI 48109-1079
bmoc@umich.edu
 Ectoparasites of vertebrates, particularly mites.

Dra. Juliana P. Sanchez
 Centro de Investigaciones y Transferencia del Noroeste de la Provincia de Buenos Aires (CITNOBA-CONICET)
 Ruta Provincial 32 Km 3.5
 2700 Pergamino
 Buenos Aires, Argentina.
julianasanchez@unnoba.edu.ar
 Ectoparasite (especially flea) systematics, taxonomy, and ecology.

Jeff Shryock, M.S.
 Sr. Research Biologist
 Merial, Ltd., Missouri Research Center
 6498 Jade Rd.
 Fulton, MO 65251
Jeffrey.shryock@merial.com

Dr. Andrew Smith
 Vet and Biomedical Sciences
 Murdoch University

Murdoch 6150
 West Australia
 Andrew.Smith@murdoch.edu.au
 Ecology; fleas as vectors of parasites and associated diseases.

Dr. Amoret P. Whitaker BSc MSc DIC DipFMS
 Scientific Associate – Forensic Entomology
 Department of Life Sciences
 Natural History Museum
 Cromwell Road London SW5 7BD
a.whitaker@nhm.ac.uk
 Forensic entomology.

Bill Wills
 Adj. Professor
 142 Heritage Village Lane
 Columbia, South Carolina 29212
pb.wills@att.net
 Flea ecology.

Fleas in Art and History

To a Louse
 On Seeing One on a Lady's Bonnet, at Church
 1786

Robert Burns (25 January 1759, Alloway, Ayrshire, Scotland to 21 July 1796, Dumfries, Scotland)

Ha! whaur ye gaun, ye crowlin ferlie?
 Your impudence protects you sairly;
 I canna say but ye strunt rarely,
 Owre gauze and lace;
 Tho', faith! I fear ye dine but sparely
 On sic a place.
 Ye ugly, creepin, blastit wonner,
 Detested, shunn'd by saunt an' sinner,
 How daur ye set your fit upon her-
 Sae fine a lady?
 Gae somewhere else and seek your dinner
 On some poor body.
 Swith! in some beggar's haffet squattle;
 There ye may creep, and sprawl, and sprattle,
 Wi' ither kindred, jumping cattle,
 In shoals and nations;
 Whaur horn nor bane ne'er daur unsettle

Your thick plantations.
 Now haud you there, ye're out o' sight,
 Below the fatt'rels, snug and tight;
 Na, faith ye yet! ye'll no be right,
 Till ye've got on it-
 The verra tapmost, tow'rin height
 O' Miss' bonnet.
 My sooth! right bauld ye set your nose out,
 As plump an' grey as ony groset:
 O for some rank, mercurial rozet,
 Or fell, red smeddum,
 I'd gie you sic a hearty dose o't,
 Wad dress your droddum.
 I wad na been surpris'd to spy
 You on an auld wife's flainen toy;
 Or aiblins some bit dubbie boy,
 On's wyliecoat;
 But Miss' fine Lunardi! fye!
 How daur ye do't?
 O Jeany, dinna toss your head,
 An' set your beauties a' abroad!
 Ye little ken what cursed speed
 The blastie's makin:
 Thae winks an' finger-ends, I dread,
 Are notice takin.
 O wad some Power the giftie gie us
 To see oursels as ithers see us!
 It wad frae mony a blunder free us,
 An' foolish notion:
 What airs in dress an' gait wad lea'e us,
 An' ev'n devotion!

**

(On the next page)

A Peasant Woman Picking Fleas off a Dog, c. 1626 –27.

Artist: Adriaen Brouwer (1605/6, Flemish, Oudenaarde to 1638 Antwerp).
 Oil painting on rectangular panel (unable to find if original was in color). Dimensions: Oval 7 1/8 x 5 3/8 in. (18.1 x 13.7 cm); panel 8 x 6 1/4 in. (20.3 x 15.9 cm).



Siphonaptera Literature (2019)

Corrigendum to: Molecular detection of vector-borne pathogens in rural dogs and associated *Ctenocephalides felis* fleas (Siphonaptera: Pulicidae) in Easter Island (Chile): (Journal of Medical Entomology (2018) DOI: 10.1093/jme/tjy141). (2019). Journal of Medical Entomology, 56(1), 297. doi:10.1093/jme/tjy198

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Obituaries

Dr. Allen H. Benton

Born: September 4, 1921, Cato, New York.

Died: September 29, 2014, Fredonia, New York.

Wed Marjorie Hall Benton 67 years, his children, Thomas and Holly Benton.

Education: BS, MS, and PhD, Cornell University, Ithaca, New York.

Military: World War II, 112th Cavalry Regimental Combat Team in Luzon (Philippines) and Japan. The regiment was the basis for Norman Mailer's "The Naked and the Dead" (1948), filmed 1958, and Francis Gwaltney's "The Day the Century Ended" (1955), filmed as "Between Heaven and Hell" (1956).

Teaching: Biology courses, SUNY (State University of New York), Fredonia, 22 years. Writer, poet, and birder.

Other works: weekly birder show (WRGB-TV); weekly newspaper columns on birds, "On the Wing", *Knickerbocker News*, and nature, "World of the Wild," *Dunkirk Observer*.

Donations go to The New York State Nature Conservancy.

His publications are listed in FN 83 December 2018.

I met Dr. Benton only once, when he gave a superb talk at a SOVE (Society of Vector Ecology) meeting in Utah on "unsolved problems of the Siphonaptera". This was published as Benton, A.H. 1982. Life strategies in the order Siphonaptera: a review of some unsolved problems. *Advances in Biosciences*, 1: 28-34.

His presentation emphasized the limited knowledge obtained about flea life-histories (still true today!), especially bat and rodent fleas. I also recall him saying that he considered his teaching successful if his student was impressed with the difficulty of establishing simply a single fact about nature.

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Dr. Alberto B. Broce

Born: January 2, 1942 in Las Tablas, Panama.

Died: November 12, 2019. Tucson, Arizona, complications of Parkinson's.

Wed 1966 to Bobara his wife of 53 years, children Andy (Terri), Bruce (Becky), and Maritza.

Education: El Zamorano Pan-American Agricultural School in Honduras, University of Florida, B.S. ('65), M.S. ('67), and Ph.D. ('71).

Teaching: Chapingo, Mexico, professor of Entomology, National Graduate School of Agriculture, 1972-73. Joined Kansas State University Entomology Department in 1979, 29 years. Livestock Entomology, Insect Behavior.

Other Work: 1973-1975 Fortín de las Flores, Lockheed Electronics; Texas A&M University at the USDA ARS Screwworm Research laboratory, NASA remote sensing technology for use in the screwworm eradication.

Donations go to the Kansas State University Foundation, Broce (Alberto & Bobara) Family Insect Zoo Excellence Fund, D54624. KSU Foundation, 1800 Kimball Ave., Suite 200, Manhattan, KS 66502.

A Celebration of Alberto's life will be held May 2, 2020, at 2:00 P.M. at College Avenue United Methodist Church, 1609 College Avenue, Manhattan, Kansas.

Dr. Broce was my doctoral major co-professor. I remember Dr. Broce fondly for his expertise and sense of humor.

Dr. Broce had at Kansas State University the "Fly Room", a name I thought ominous, containing stacks and stacks of cubic-foot, screened cages. There, many flies of medical, veterinary, and livestock significance were reared, including face, horn, house, and stable flies.

One of the many interesting researches he conducted was a scanning electron microscopy study comparing prestomal teeth in different flies. Some (house flies) have sponging mouthparts, some (face flies, screwworm flies) rasping mouthparts, and still others (stable flies) have piercing mouthparts (Broce, A.B., Elzinga, R.J. Comparison of prestomal teeth in the face fly (*Musca autumnalis*) and the house fly (*Musca domestica*) (Diptera: Muscidae). (1984) Journal of medical entomology, 21 (1), pp. 82-85).

Dr. Broce's research was on Diptera primarily, but his work on trapping, insecticide resistance, and behavior of Siphonaptera is notable.

Partial list of Alberto Broce's publications

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